

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | **Context:** South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU’s 57-subject taxonomy is organized around US/Western academic disciplines [\[Q1, Q9, Q18\]](#) and contains no subject addressing agro-ecology, delta hydrology, haor wetland systems, boro/aman/aus rice cultivation, char-land farming, Deccan dry-land cropping, or coastal aquaculture — the precise subdomains required for this deployment. The dataset analysis confirmed this empirically across 175+ examples sampled from 37 configs: the closest environmental content observed is generic water chemistry and pastoralism definitions, with no South Asian agricultural systems represented [\[DATASET-D19, D22, D32\]](#). Many subjects are actively irrelevant (US foreign policy, US history, US government), introducing construct-irrelevant variance for an agricultural science deployment. Given IO is a HIGH-priority dimension and the gap is structural and complete, this is a major validity violation.

ID	Evidence
Q1	“We propose a new test to measure a text model’s multitask accuracy. The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more.” (p.1)
Q9	“The benchmark covers 57 subjects across STEM, the humanities, the social sciences, and more. It ranges in difficulty from an elementary level to an advanced professional level, and it tests both world knowledge and problem solving ability.” (p.1)
Q18	“We create a massive multitask test consisting of multiple-choice questions from various branches of knowledge. The test spans subjects in the humanities, social sciences, hard sciences, and other areas that are important for some people to learn. There are 57 tasks in total, which is also the number of Atari games (Bellemare et al., 2013), all of which are listed in Appendix B.” (p.3)
D19	The way of life based on breeding and herding of animals that is used as a source of food, shelter, and clothing is called pastoralism — Generic geography — no delta/wetland/aquaculture content
D22	In fungi, cell walls are reinforced by which of the following? (A) Chitin (B) Silica (C) Lignin (D) Collagen fibers — General biology — not agricultural/agro-ecological
D32	A substance that is NOT generally considered to be a toxic pollutant in water is (A) carbonic acid (B) a halogenated hydrocarbon (C) lead (D) mercury — Closest approach to environmental science — but generic water chemistry, not delta/haor/saline intrusion relevant

Strengths

- Universal STEM reasoning items (abstract algebra, formal logic, physics) provide a culturally-neutral baseline for cross-model comparison [\[DATASET Strength 1\]](#)
- Statistical inference items (hypothesis testing, sampling, Type I/II error) test methodology skills transferable to environmental science research [\[DATASET-D30\]](#)

ID	Evidence
D30	The Department of Health plans to test the lead level in a specific park. Because a high lead level is harmful to children, the park will be closed if the lead level exceeds the allowed limit. — Environmental testing question, but no South Asian or agricultural context

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for this deployment include haor/beel wetland ecology, boro/aman/aus rice cultivation, char-land farming, Brahmaputra-Jamuna floodplain hydrology, Deccan Rabi/Kharif dry-land cropping, brackishwater shrimp aquaculture, and Bangladesh-India transboundary water management — none of which are present in the 57-subject taxonomy.

IO-2: MMLU's taxonomy omits every regionally-relevant agricultural and environmental science category. Subjects are bounded by US/Western academic disciplines [Q9, Q18, Q33], with history/geography framed through Western lenses [Q28, DATASET-D16, D20]. The 'breadth' claim of 57 tasks [Q11, Q83] does not translate into agro-ecological coverage.

IO-3: MMLU includes substantial content irrelevant to South Asian agriculture: US Foreign Policy [Q171, DATASET-D5, D6], US History [Q151, DATASET-D2], US Government [DATASET-D3, D4], US tort/contract law [DATASET-D7, D8, D25], and Western European history [DATASET-D16, D17]. These consume evaluation weight without informing agricultural reasoning.

IO-4: The complete absence of any agro-ecological subject combined with heavy weighting toward US-centric subjects creates content underrepresentation (missing constructs) and construct-irrelevant variance simultaneously — both validity-harming patterns.

ID	Evidence
Q1	"We propose a new test to measure a text model's multitask accuracy. The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more." (p.1)
Q9	"The benchmark covers 57 subjects across STEM, the humanities, the social sciences, and more. It ranges in difficulty from an elementary level to an advanced professional level, and it tests both world knowledge and problem solving ability." (p.1)
Q11	"Since our test consists in 57 tasks, it can be used to analyze aggregate properties of models across tasks and to track important shortcomings." (p.2)
Q18	"We create a massive multitask test consisting of multiple-choice questions from various branches of knowledge. The test spans subjects in the humanities, social sciences, hard sciences, and other areas that are important for some people to learn. There are 57 tasks in total, which is also the number of Atari games (Bellemare et al., 2013), all of which are listed in Appendix B." (p.3)
Q28	"Finally, our history questions cover a wide range of time periods and geographical locations, including prehistory and other advanced subjects." (p.4)
Q33	"STEM subjects include physics, computer science, mathematics, and more." (p.4)
Q83	"By covering 57 subjects at varying levels of difficulty, the test assesses language understanding in greater breadth and depth than previous benchmarks." (p.8)
Q117	"Table 2: Summary of all 57 tasks." (p.15)
Q151	"Figure 44: A High School US History example." (p.21)
Q171	"Figure 68: A US Foreign Policy example." (p.26)
D2	"The song, and the movement that it was connected to, highlight which of the following developments in the broader society in the late 1800s?" (re: "The Hayseed", US Populist movement) — US-specific historical content with no South Asian relevance
D3	The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — Entirely US constitutional law content
D4	Both the War Powers Act of 1974 and the Budget and Impoundment Control Act of 1974 represent efforts by Congress to limit the powers of the president. — US-specific legislative history
D5	In American government, the power to declare war rests with Congress. — US government structure; no relevance to South Asian governance or environmental policy
D6	Why is NSC 68 seen as a turning point in US Cold War Foreign policy? — US Cold War policy — no South Asian agro-ecological relevance

D7	A truck driver was assigned to drive a huge tractor-trailer loaded with logs. The logs were being sold by a lumber company... the girl was struck by the tractor-trailer and was seriously injured. — US tort law scenario — US legal context
D8	An avid baseball fan learned that a local professional baseball club was conducting a baseball camp... forwarded the club a \$50 registration deposit — US contract law — US sports/cultural context
D16	The Scribbling-Machines have thrown thousands of your petitioners out of employ... Leeds Woolen Workers Petition, 1786 — Entirely European industrial history — no South Asian relevance
D17	What is tolerance?... Of all religions, the Christian ought doubtless to inspire the most tolerance... Voltaire, Letters on the English Nation, 1733 — European Enlightenment philosophy — no South Asian agricultural relevance
D20	Which zone contains low-income slums, ethnic ghettos, and general deterioration in Burgess's concentric **zone model of urban form***? — US urban planning model (Burgess model, Chicago School)
D25	Bill purchased a can of Sipep from the Ajax Minimart. After he finished drinking the Sipep, Bill noticed that the can contained dead insects stuck on the inside bottom of the can. In a strict product liability tort action against Ajax, Bill must prove... — US-style product liability tort — US legal culture

Information Gaps

- Approximately 20 of 57 configs were not directly sampled by the dataset analysis; rare South Asian agricultural items in unreviewed configs cannot be entirely ruled out, though structural sourcing makes them improbable.

Requires Expert Verification

- Whether any single MMLU item across the full 14,079-question test set addresses South Asian agro-ecology would require exhaustive scan of all 57 configs by a regional expert.

Remediation

Gap 1: MMLU's 57-subject taxonomy contains no agro-ecological, delta-ecology, or South Asian regional agricultural subjects — the precise constructs the deployment must measure.

Recommendation: Treat MMLU as a generic STEM/reasoning baseline only and pair it with a purpose-built South Asian agricultural knowledge benchmark organized by AEZ (haor, char-land, Deccan dry-land, coastal aquaculture) co-designed with BAU, BRRI, PJTSAU, and ANGRAU. Restrict MMLU reporting to a curated subset of culturally-neutral STEM configs (abstract algebra, college mathematics, college physics, formal logic, high school statistics).